RAIL TRANSIT SEMINAR • SEPT 2-3, 2026



EXPERIENCES WITH GRINDING AND MILLING IN NORTH AMERICA – A SUPPLIER'S VIEW

BOSTON, MA



Refresher – Grinding and Milling

Grinding and Milling - Basics

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Rail Grinding

- Abrasive (dry) process with rotating grinding stones
- Face grinding technology: Rotational axis perpendicular to rail surface



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Rail Milling

- Non – abrasive, dry cutting process with rotating cutter heads (carbides)
- Peripheral milling: Rotational axis parallel to rail surface



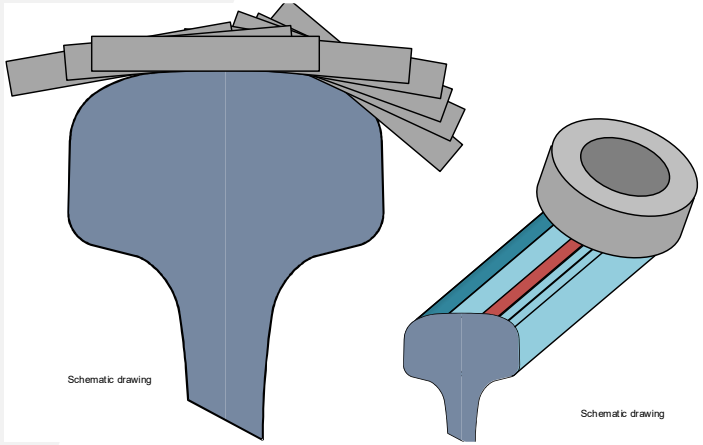


Grinding and Milling: Transversal Profile

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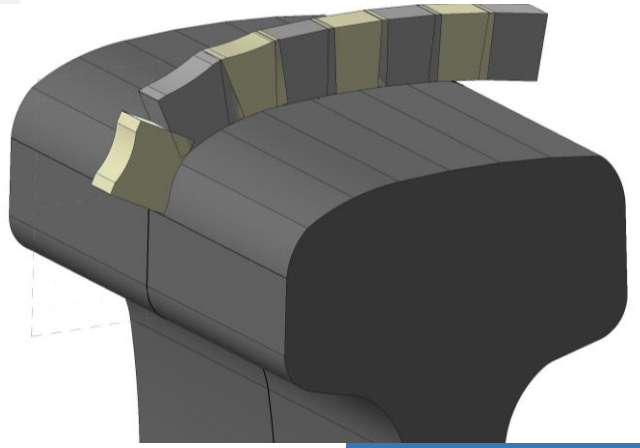
Rail Grinding:

- Flexible profile adjustment through angular stone alignment



Rail Milling:

- Profile defined by negative shape of cutter head (milling tool) with carbide inserts



Grinding and Milling - Results

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Resultant product looks similar, but milling ≠ grinding







The Supplier's Journey

The Starry-Eyed Supplier

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- North America: Rail milling white spot
- Rail Milling successful in Europe, Asia and to some extend Australia
- Europe: 165,000 mi
- USA + CAN: 177,000 mi
- Extrapolation for milling machine potential





The Best Invention Since Sliced Bread

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- Common idiom used to praise an invention or development
- A few words about “Sliced Bread”
- Bread slicing machine invented 1912 by Otto Frederick Rohwedder in Davenport/Iowa
- Took 16 years to get traction (1928)



The Best Invention Since Sliced Bread

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- Corrective / regenerative maintenance
  - Remove a lot of metal in one pass
- High quality rail surface
- Spark and dust free
- By-product collection
- Implemented in Europe and Asia

- Only preventive grinding
- Milling removes too much metal
- Milling is sooo slow
- We have 4 different target profiles
- Happy with current quality
- What's the value of dust free

The Sobered Supplier

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- Rail is Rail, but there are significant differences between continents
  - Technical, legislative, economical and cultural
- Grinding is engrained as maintenance technology
  - The only technology for 50+ years
  - Some technical barriers
  - Monopolistic (grinding) situation



The Educating Supplier

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Challenge: Would you trust your supplier educating you about his products?











The Solution to the Challenges

Milling Removes too Much Metal

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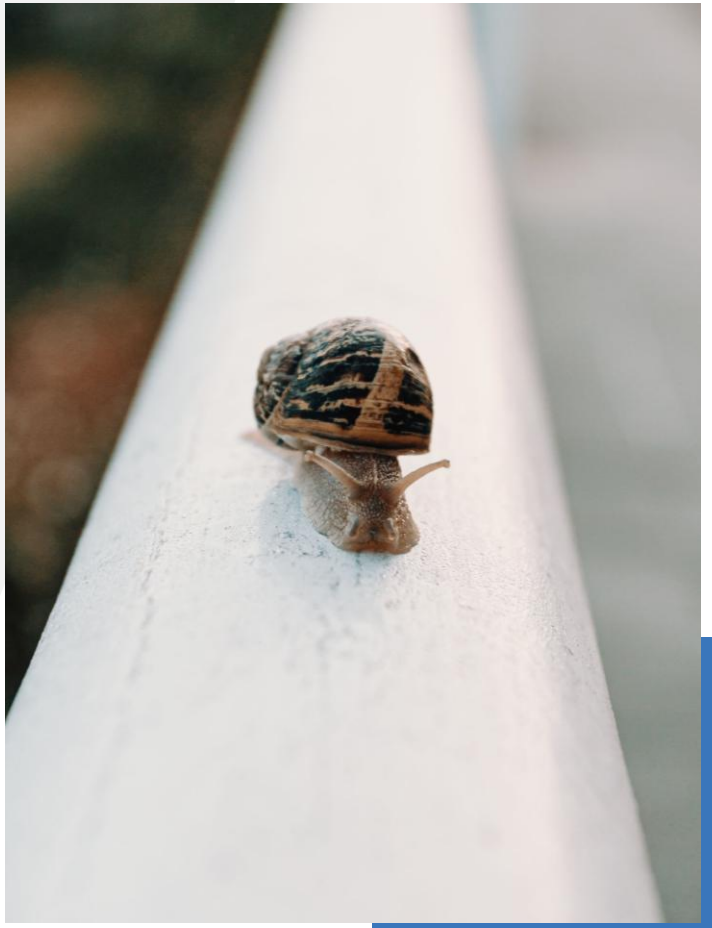
- Milling can remove 0.3mm or more per pass
- Max. cutting depth (per pass):
  - Machine dependent 1.0 – 6 mm
  - Transit (small machine): 1-1.5mm
  - Controllable in 0.1mm steps
- Theoretically min. cutting depth of 0.1mm
  - Challenging due to elasticities of track and machine



Milling is Soooo Slow

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- Two different speeds
  - Speed per pass (ft/h, mi/h – m/h, km/h)
  - Finished track per hour (ft/h, mi/h – m/h, km/h)
- Speed per pass
  - Transit milling machines: 328ft/h – 3937ft/h (100m/h – 1200m/h)
- Finished track per hour
  - Typically lower than speed per pass



Speed per Pass Overview

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- Several milling machines currently operating in North America



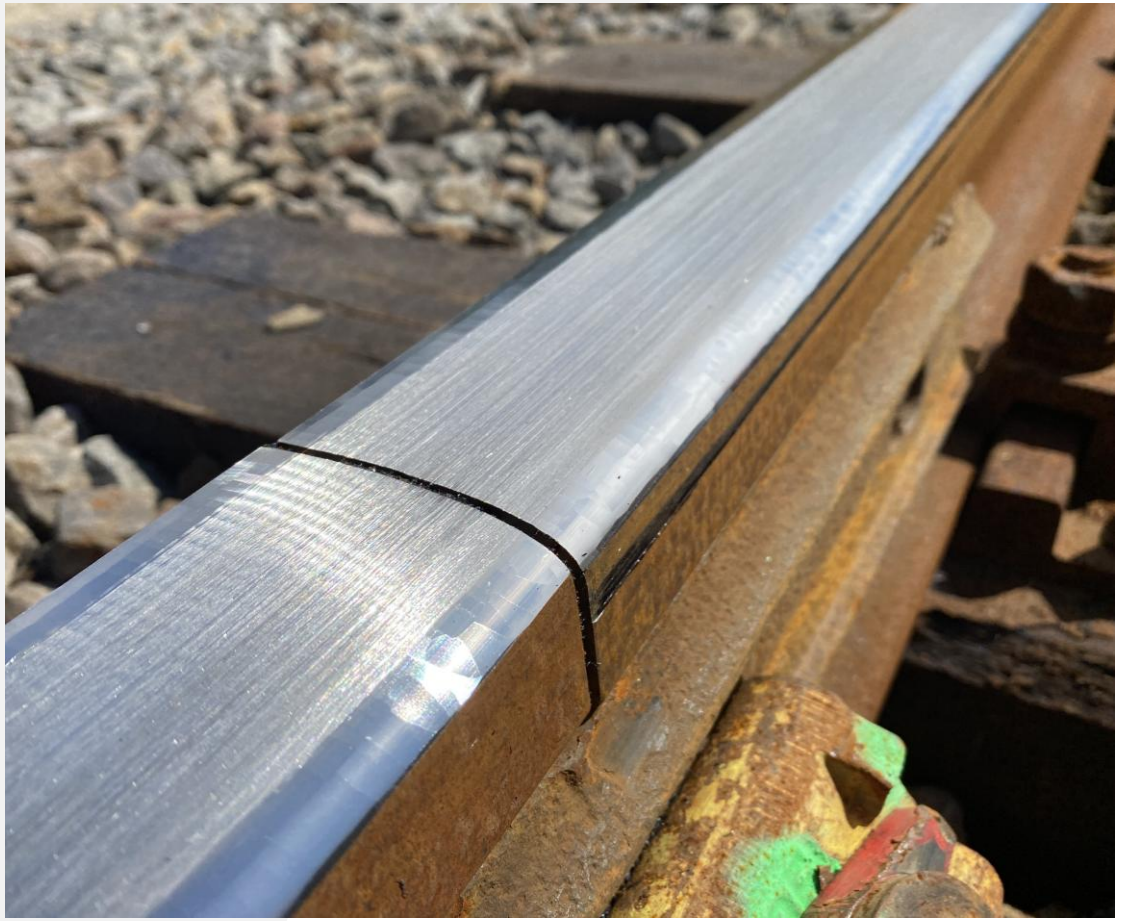
- Individual milling machine speed range depends on milling machine type and milling technology

*According to machine flyers and supplier / OEM websites

Finished track per hour

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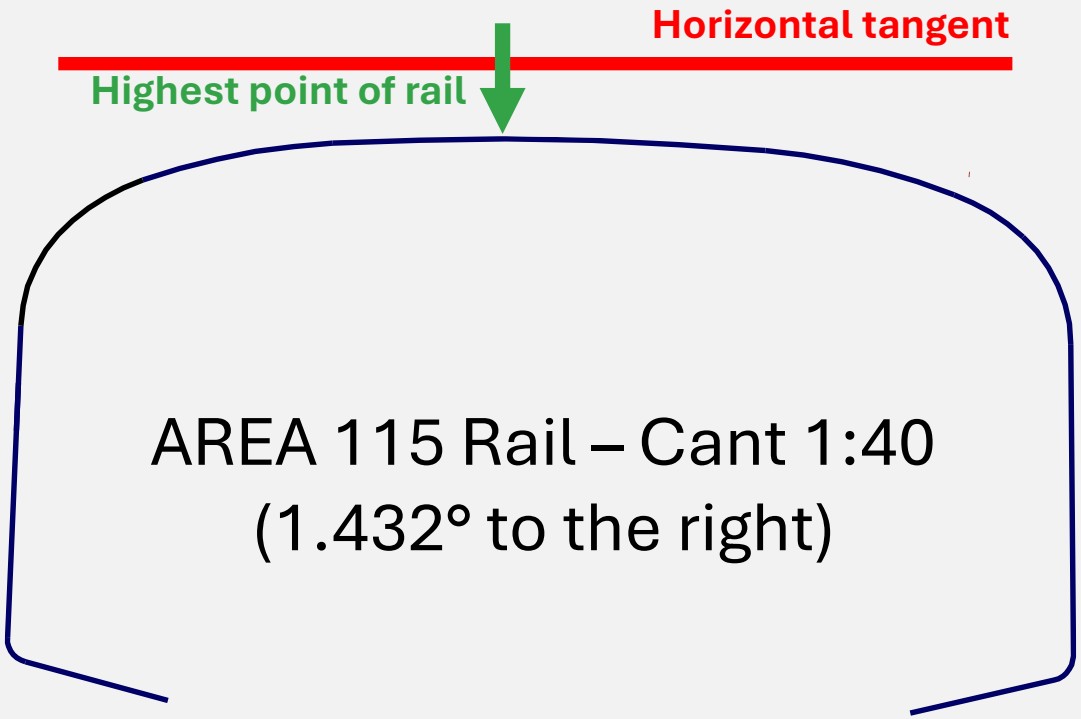
- Processing speed per pass
  - Milling machine and technology
  - Time for ramp-in and ramp-out
- Required metal removal per pass and number of passes
  - Initial profile shape
  - Surface damage condition
- The more metal to remove per pass (area) the slower the pass



Metal removal (per pass)

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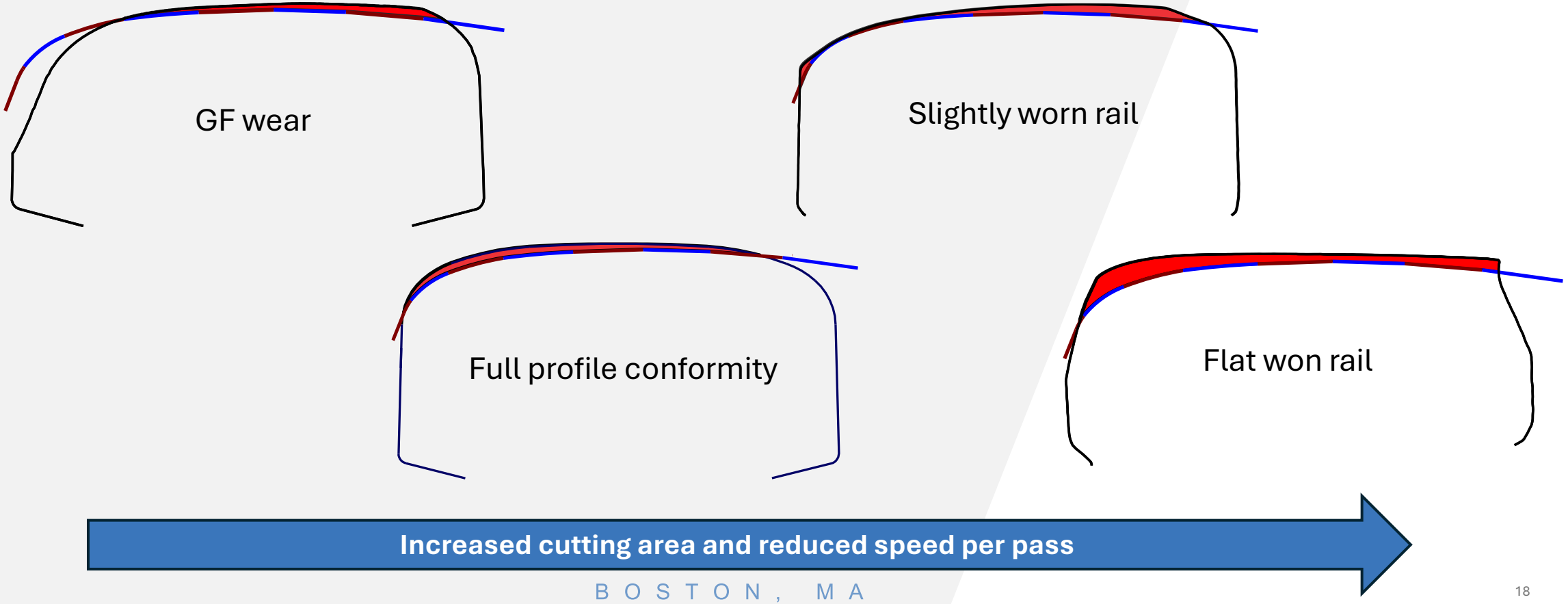
- Material removal in reference to highest point of rail (HPR)



Cutting Depth vs. Area Removed

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- Examples: Same cutting depth – different area (red) removed



Metal Removal and Grinding

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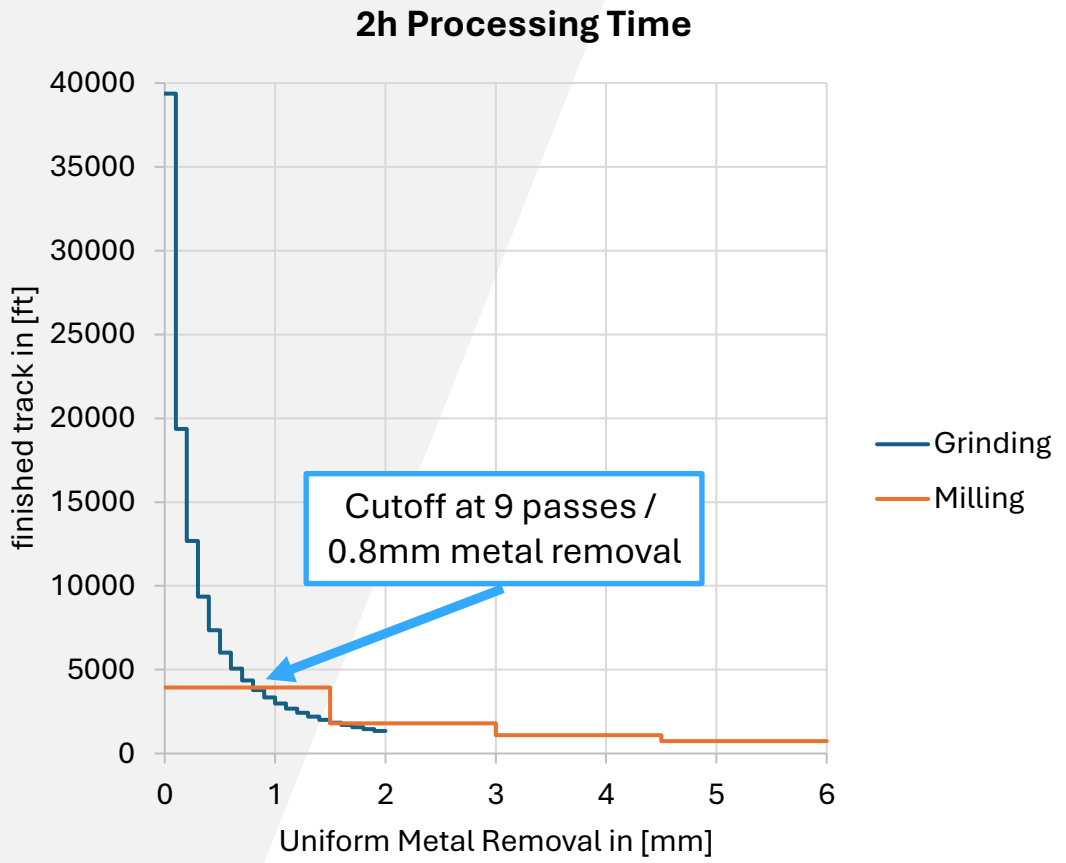
- Grinding stones aligned across transversal profile
  - Profile created as polygon
- 12 stones per rail required to cover full profile in one pass
  - Dependent on acceptable facet width for final pass
- Metal removal per pass
  - Uniform metal removal across full profile vs. localized higher removal



Example: Finished track in 2h

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- 2 h of total processing time window
- Uniform Metal removal
  - Grinding 3.73 mph and 0.1mm
    - 12 stones per rail for full profile
  - Milling 0.373 mph and 1.5mm
- Grinding: 2 min for directional change
- Milling: 10 min to set back
- Cut-off: 9 passes / 0.8mm

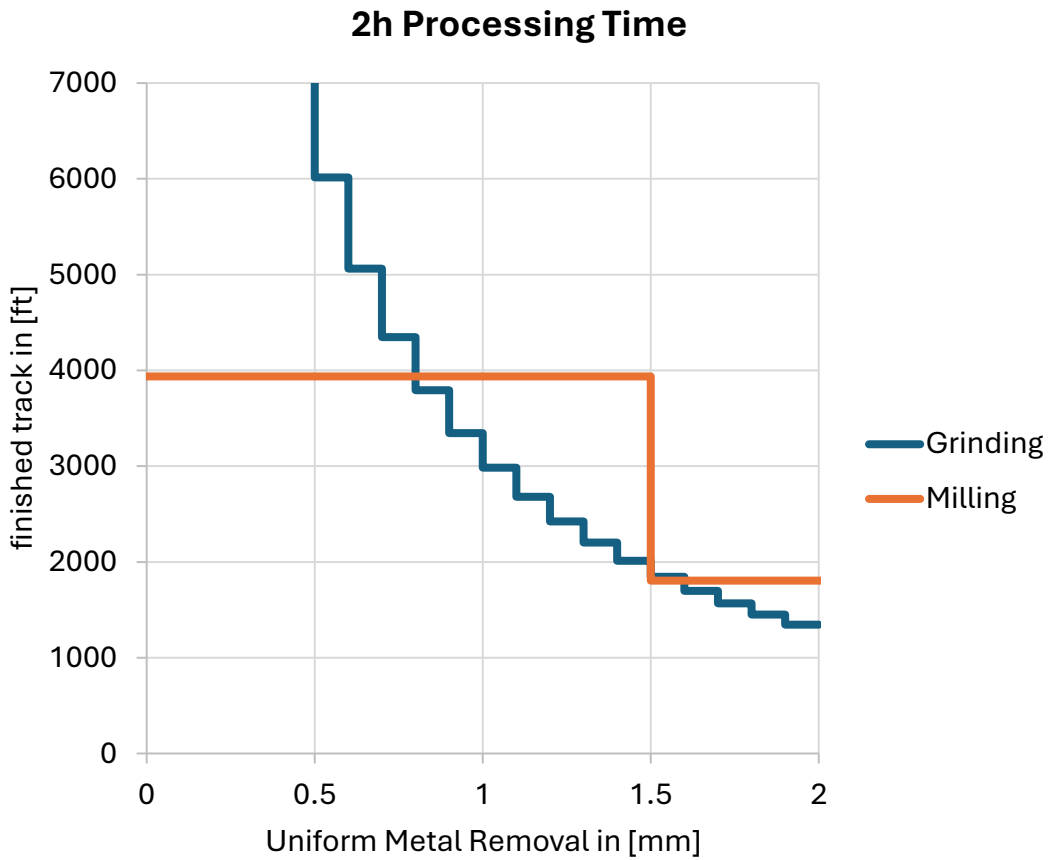


Simplified, schematic diagram

Example: Finished track in 2h

Plasser American

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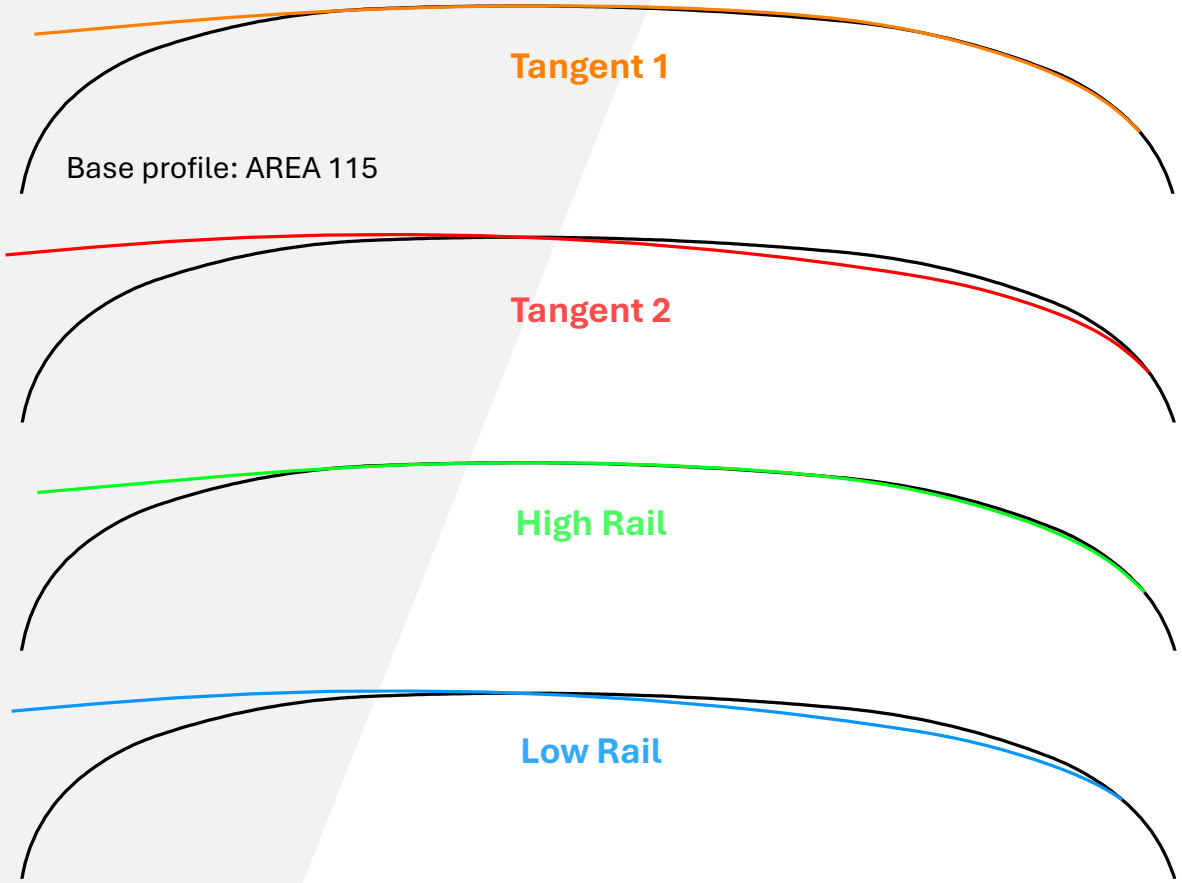


Simplified, schematic diagram

Target Profiles

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- Significant difference between Europe (1-2 profiles) and North America (multiple)
- Transits typically have 4 target profiles
  - 2 tangent profiles – distribute wheel wear over a greater area
    - 50% - 50%
  - One high rail and one low rail profile



Example Profiles

Target Profiles and Milling

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- Every profile needs an according pair of cutter heads
- Change cutter heads to change to a different profile
  - Cassettes or full cutter head
- Cutter head change: 10-15 min
  - Out of 2h milling time: 8-12%



Target Profiles Creation

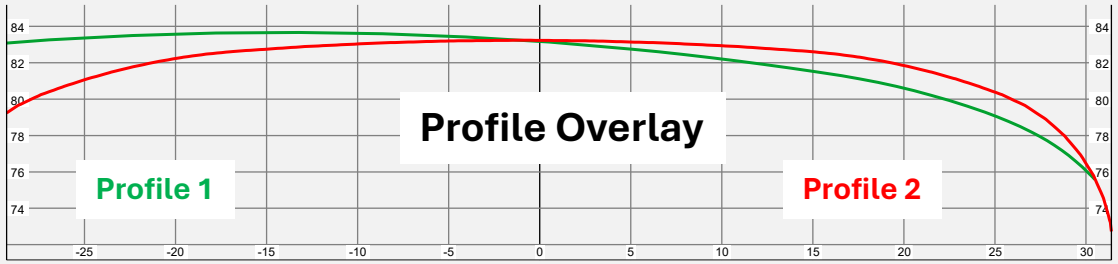
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- Target profiles vs. creation accuracy
  - Best case: ± 0.3mm for grinding, ± 0.2mm for milling
  - Specification?
- Specification vs. quality control
  - Star Gauge or Bar Gauge limited measurement precision
  - Electronic measurement technology

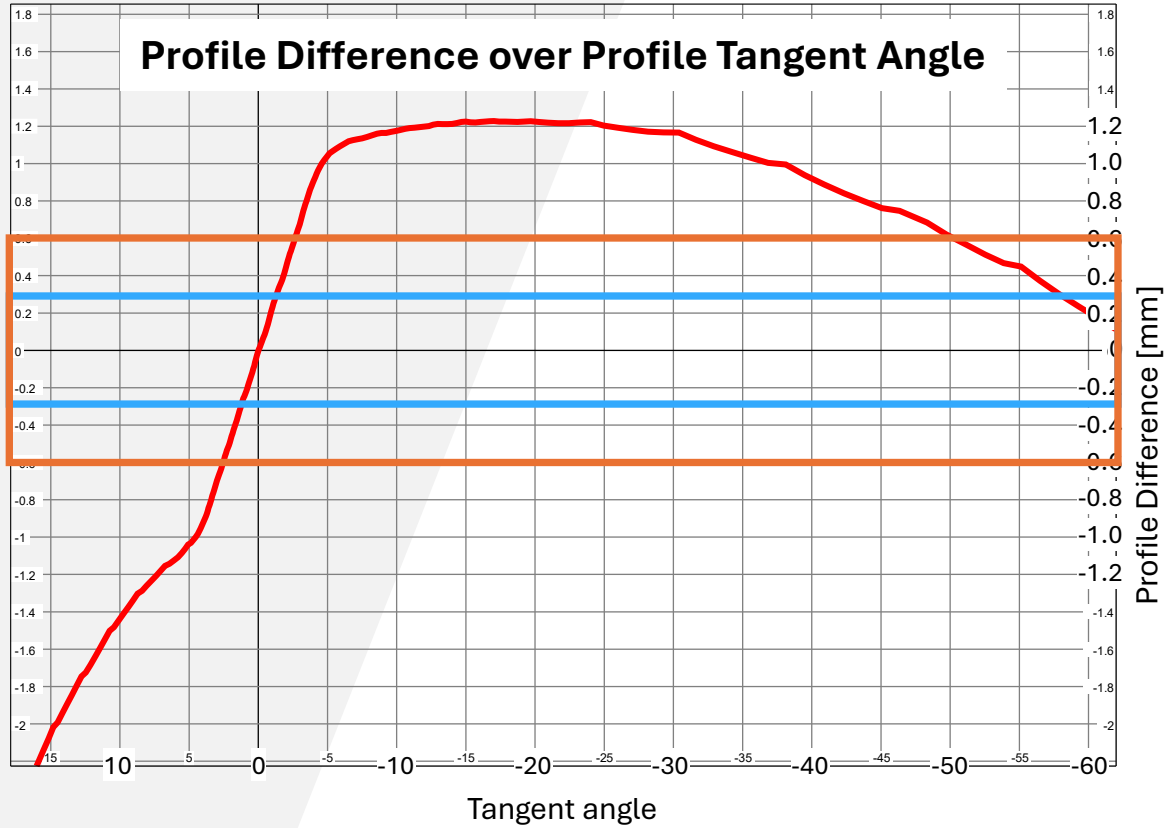


Example: Significant Profile Difference

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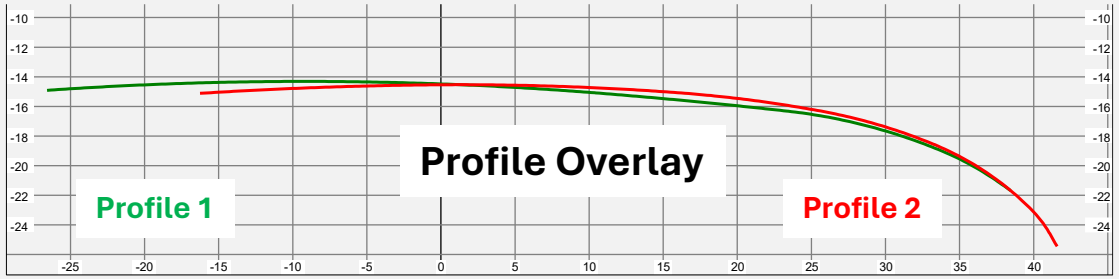


- 2 target profiles with significant difference
- Profiling Specifications
  - +/- 0.3mm ok
  - +/- 0.6mm ok

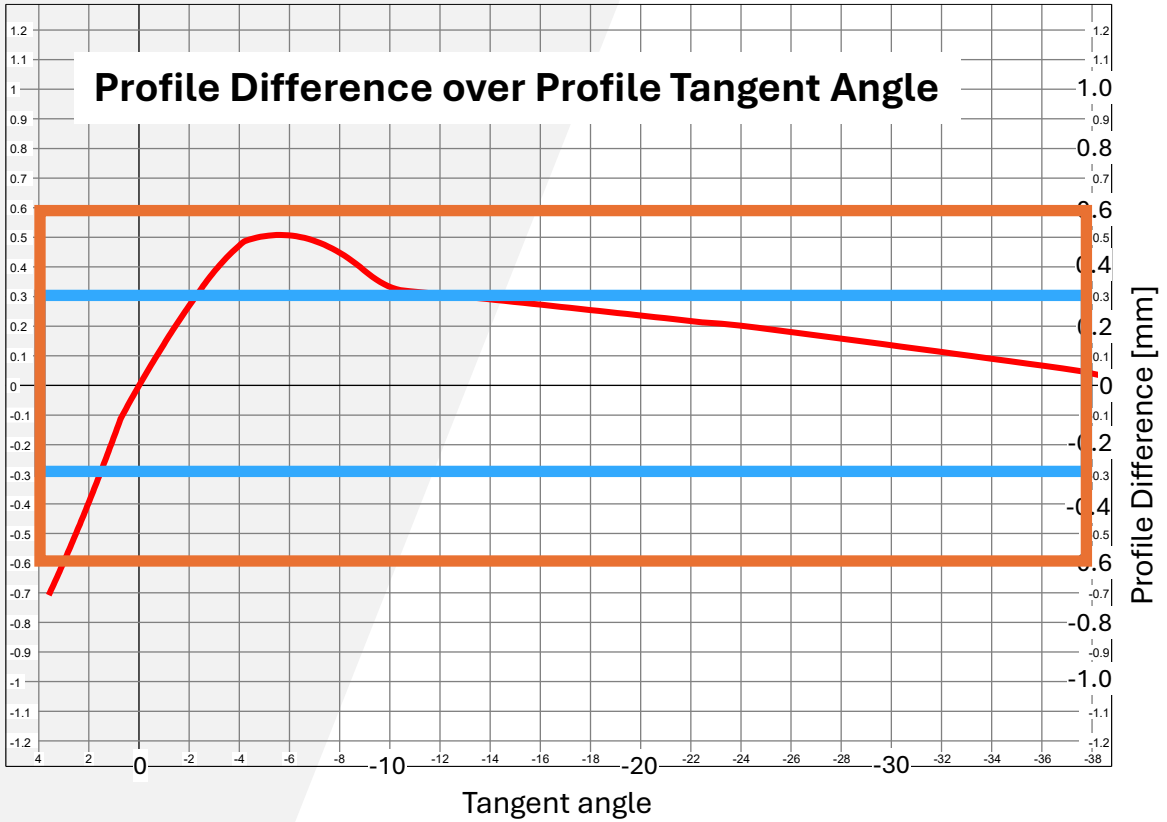


Example: Medium Profile Difference

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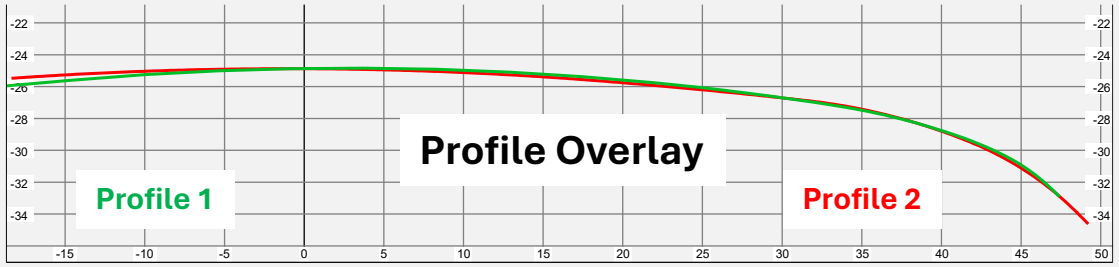


- 2 target profiles with medium profile difference
- Profiling Specifications
  - +/- 0.3mm ok
  - +/- 0.6mm not feasible

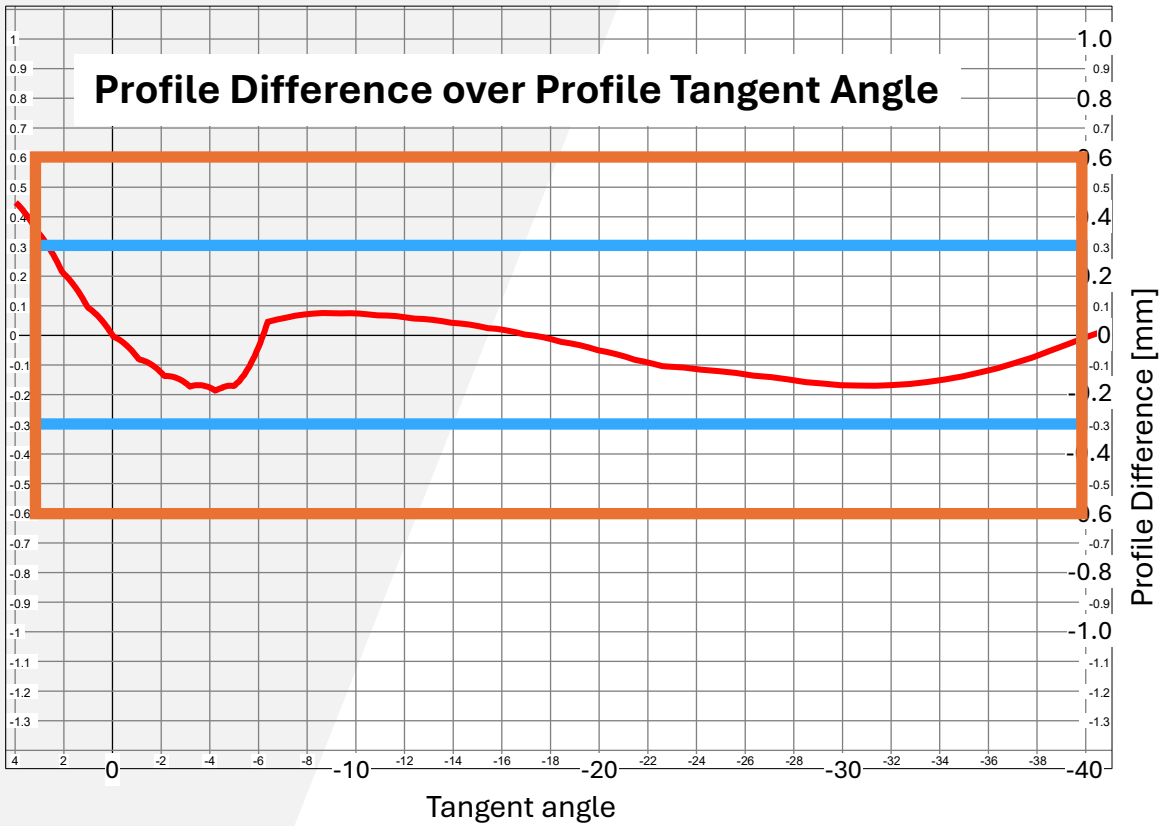


Example: Minor Profile Difference

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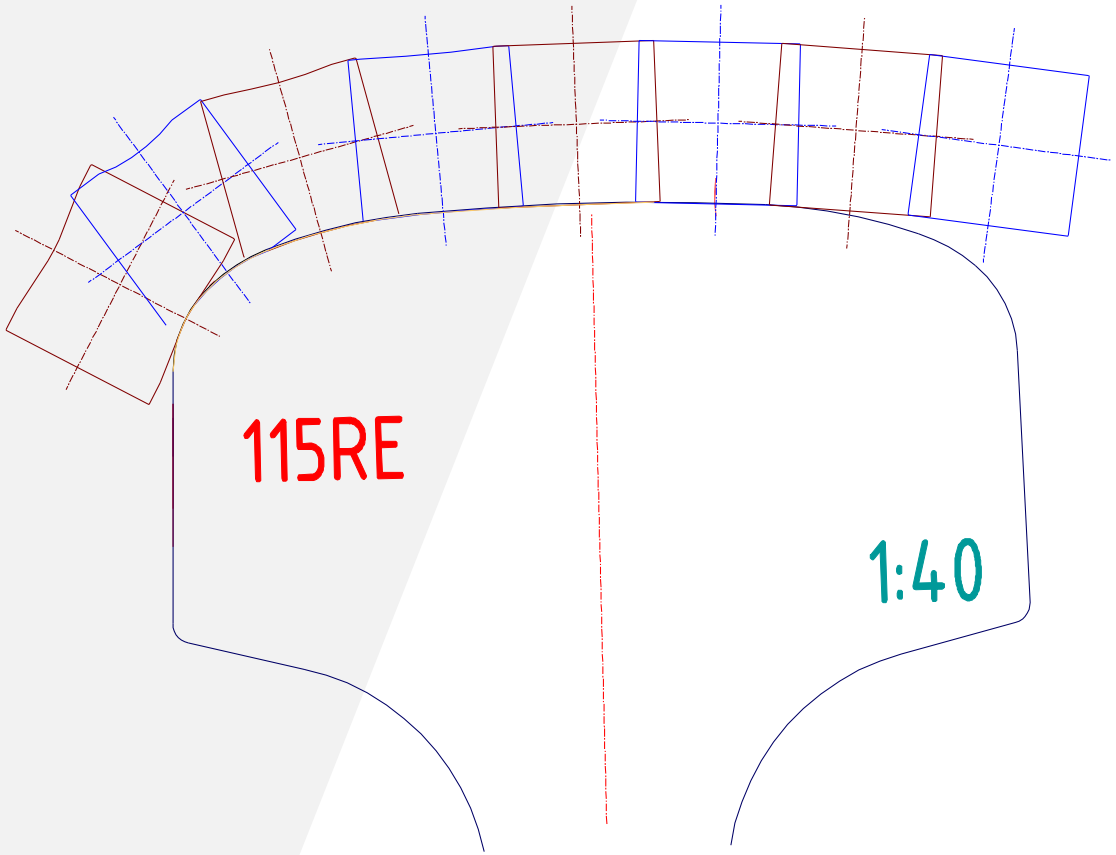
- 2 profiles with minor profile difference
- Profiling Specifications
  - +/- 0.3mm not feasible
  - +/- 0.6mm not feasible



Target Profiles and Milling

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- Focus/strength of rail milling:
  - Removal of (deep) damage
  - Correction of severely deformed profiles
- Milling: 1-2 target profiles
  - Benefit from damage removal higher than getting the profile 100% right
  - Additional benefits
  - Maximized Productivity
- Potentially: touch up with grinding at a later stage



Additional Benefits and their Value

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- Dust free / low dust
  - Only polishing may create limited dust – collection system
  - Air quality in tunnels, stations, next to the machine
- Spark free / low spark
  - Minimal fire risk
  - No damage to nearby objects
- Battery drive
    - Zero Exhaust / NOx / CO
- Challenge: $$$ Value



Technology Specific Requirements

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- Different requirements for milling and grinding
- Grinding:
  - Spark and Dust-handling, stone change, stabling location and access to machine
- Milling:
  - cutter head / cassette change, chip-bunker evacuation, milling insert indexing, work-shop container, stabling location and access to machine







How to Translate This Into RFQs

The Technical Starting Point

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- What is your current track condition
  - If you do not know, get support
  - Independent, knowledgeable, reliable third party that you trust
- What is the required work
  - A shiny rail
  - Preventive work
  - Corrective work
  - Regenerative work



Technical RFQ Compilation

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- Use previous RFQs as starting point
  - Do not copy and paste without review
- Specific technology vs technology open: milling and/or grinding
- If you do not have in-house capability, get support.
  - Independent, knowledgeable, reliable third party that you trust



Technical Work Requirements

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- Clearly define your requirements
  - Type of maintenance
  - How many shifts
  - Target profiles, accuracy, surface quality and quality control
  - Additional requirements
- Provide as much information about your track condition as possible



Tender Review Process

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- Price vs. Budget
- Not just price but what gets done with this money
  - Machine performance vs. price
- Definition of machine performance
  - Finished track/h
  - The clearer your tender definition the easier the comparison



Full Circle: Finished track per h

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- Influencing Factors
  - Technology
  - Metal Removal + Definition
  - Speed per pass
  - No. of passes
- According to RFQ Specification
  - Transversal and long. profile, profile accuracy, surface quality, damage removal
  - Quality Control and measurement precision
- Comparability

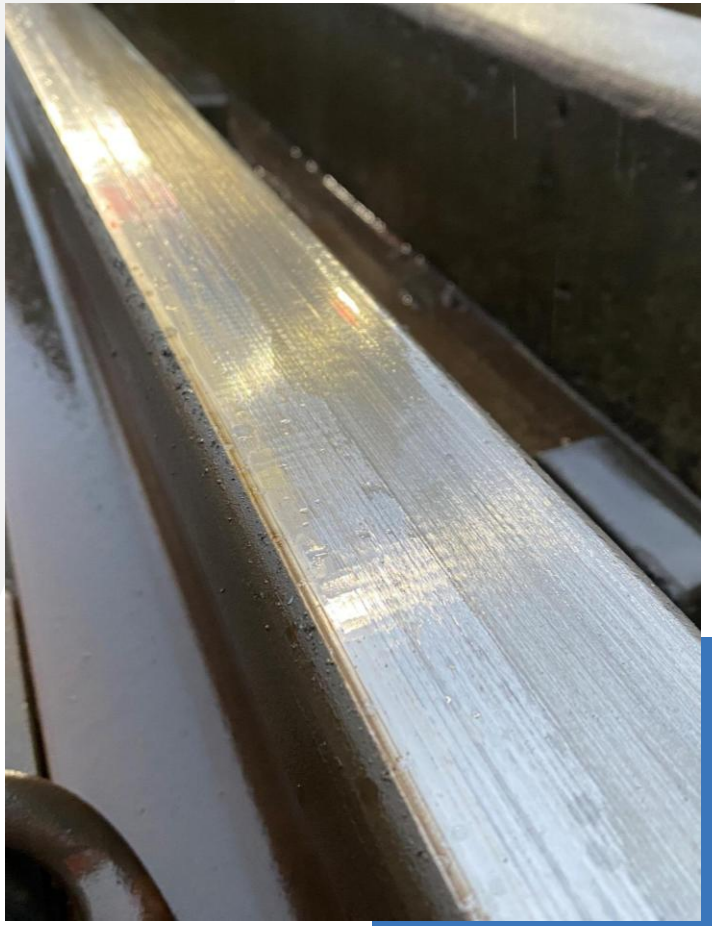


Well Maintained Rail

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Summary

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- A Supplier's View
  - Milling and Grinding
- Importance of finished track per hour
  - Technology specific influencing factors
  - According to Specifications
- Importance of well-defined Scope of Work in RFQ





Thank You for Your Attention

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